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Dynamic responses to smoking bans: Evidence from young adults in a developing country☆

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ABSTRACT

Smoking bans have been widely implemented, despite mixed evidence on their effectiveness in reducing smoking prevalence. This paper provides novel insights into the dynamic impacts of smoking bans in the context of Brazil, a large developing country with 18.4 million regular smokers in 2013. Our estimation strategy exploits the staggered implementation of comprehensive smoking bans in Brazilian state capitals using an event-study framework. We also leverage the variation in policy enforcement across cities. Our results indicate that bans reduced smoking prevalence by 18% among young adults, particularly when rigorously enforced. This effect is primarily driven by smoking cessation, while the impact on initiation is relatively modest. This policy change prevented roughly USD 87 million in healthcare costs in the treated cities.

1. Introduction

Tobacco use is the leading cause of preventable deaths worldwide, associated with various life-threatening diseases, including cancer, heart disease, and respiratory conditions. As a result, smoking exerts a significant burden on healthcare systems, and can negatively impact child development and the accumulation of human capital by diverting resources from essential needs, particularly in low- and middle-income countries (LMICs).[1] Within this context, public health mandates may play an important role in preventing smoking addiction and its associated adverse effects. Specifically, smoke-free environments might reduce exposure to passive smoking in public places, raise the marginal cost of smoking, and reshape social norms and attitudes toward smoking by limiting its visibility.

While smoking bans in public indoor spaces are widely adopted, their impact on reducing smoking prevalence remains a subject of debate (DeCicca et al., 2022). Many European studies find limited or null effects on prevalence, likely due to short-term exposure to partial

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[1] See, for example, WHO (2008, 2019), and DeCicca et al. (2022) for reviews on the health effects of smoking, and Goodchild et al. (2018) for the costs imposed to health systems. Block and Webb (2009) show that expenditure on addictive goods might lead to child malnutrition among low-income families. Harris et al. (2015) found that anti-smoking regulations in Uruguay effectively encouraged cessation among pregnant women, leading to improved birth weight outcomes. Heckman (2007) and Hai and Heckman (2022) study the relationship between health/addiction and human capital accumulation.

[2] European smoking bans typically permit designated smoking areas. In the U.S., many anti-smoking policies are determined at the state level, which may result in divergent smoking trends across states, potentially confounding the effects of smoking bans (Adda and Cornaglia, 2010). A recent review by DeCicca et al. (2022) highlights the challenge of reconciling mixed findings in the literature and the consensus to evaluate comprehensive bans. By 2020, 70% of countries had introduced smoke-free environments, but only half of these bans were comprehensive (WHO, 2021).

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restrictions (e.g., Anger et al., 2011; Jones et al., 2015; Boes et al., 2015). Effects are also inconclusive for the U.S., even in studies that exploit longer-term variation across states (Adda and Cornaglia, 2010; Carton et al., 2016).[2] Our paper contributes to this debate by offering novel evidence on the effects of comprehensive bans (i.e., enclosed public places being entirely smoke-free) in Brazil. With 18.4 million regular smokers in 2013, Brazil has implemented increasingly strict tobacco control policies, earning recognition as a global leader in this area (WHO, 2019).

Our study leverages the staggered introduction of bans across states and state capitals between 2009 and 2013, independently of other tobacco control measures. We analyze the smoking trajectories of young adults (15 to 29 years old), applying a clear and robust estimation approach that incorporates recent advances in difference-in-differences methodology (Sun and Abraham, 2021; Goodman-Bacon, 2021; Wooldridge, 2021).[3] Our main results show that local smoking bans led to a 18% reduction in smoking prevalence after four years of exposure to the policy, with delayed but increasing effects over time. This could be due to delays in enforcement, as well as the addictive nature of smoking, which requires time to adjust. We further explore the underlying mechanisms by showing that the reduction in smoking prevalence is mainly attributed to cessation among young adults who had started smoking shortly before the bans were introduced. Overall, cessation accounts for nearly 60% of the effects. The remaining portion can be attributed to preventing smoking initiation.

We are among the first to study the effects of smoking bans in the context of LMICs, which bear a disproportionate share of smoking-related costs (Goodchild et al., 2018; WHO, 2021). Anti-smoking policies in LMICs remain underexplored in the literature, and their implementation presents unique challenges, often distinct from those in high-income settings. For instance, fiscal measures such as increasing cigarette taxes may impose a heavier financial burden on poorer households, without significantly reducing smoking prevalence due to low price sensitivity. Furthermore, the impact of such policies may be less pronounced in LMICs due to the widespread availability of illicit markets, which offer cheaper, untaxed, and often more harmful alternatives (Joossens et al., 2010; World Bank, 2019).[4] Our findings support comprehensive smoking bans as an effective policy in these contexts.

In a related study, Catalano and Gilleskie (2021) found that bans in Argentina reduced smoking prevalence, particularly among younger males, attributing these effects to reduced initiation. In our study, thanks to the availability of individual-level smoking trajectories from the 2013 National Health Survey (PNS), we are able to more systematically decompose cessation and initiation patterns.[5] We find that bans can indeed operate through reducing initiation, but that early cessation is an important channel in the context of Brazil. Although the effects on initiation exhibit a clear declining trend, they are modest and less precisely estimated compared to the effects on cessation. One potential explanation is that individuals in our sample might be introduced to smoking in non-targeted environments, as initiation typically happens before the minimum legal age to purchase cigarettes and alcohol.[6]

Our focus on young adults is motivated by two key arguments. First, smoking bans might influence the likelihood of initiation – which typically happens at early ages – by reducing exposure to triggers such as peer influences during social gatherings. Second, these policies can directly raise the marginal cost of smoking by limiting access to smoking areas and reducing social acceptance, thereby encouraging cessation. Previous studies analyzing cessation patterns typically use an unrestricted sample of (older) adults. However, given the addictive nature of smoking, we argue that cessation effects may take time to materialize, potentially explaining the mixed findings in the literature. In contrast, during the early stages of addiction, the increased costs associated with smoking bans may be sufficient to accelerate the process of quitting.[7]

This paper makes valuable contributions to the literature on tobacco control policy evaluation by demonstrating the importance of dynamics and adjustment periods in shaping addictive behaviors. We further advance the broader discussion on policy design, implementation, and evaluation in contexts with limited institutional capacity. Using data from local monitoring agencies on penalties for policy violations, we examine the differential effects of smoking bans depending on enforcement levels. Our results highlight that enforcement is critical for the effectiveness of smoking bans in LMICs. Specifically, rigorously enforced bans produced effects comparable to those observed in the U.S. (Carton et al., 2016) and Europe (Pfeifer et al., 2020), while weakly enforced bans had no significant impact. Hallward-Driemeier and Pritchett (2015, p. 123) synthesize the challenges of regulation in developing countries:

> When strict *de jure* regulation and high rates of taxation meet weak governmental capabilities for implementation and enforcement, we argue that researchers and policymakers should stop thinking about regulations as creating "rules" to be followed, but rather as creating a space in which "deals" of various kinds are possible.

Finally, we also contribute to the discussion on whether smoking bans can mitigate the negative externalities of passive smoking, one of the primary justifications for their implementation (e.g., Kvasnicka et al., 2018; DeCicca et al., 2022). Adda and Cornaglia (2010) note that the net effects on passive smokers are unclear, depending not only on the direct reductions of cigarette consumption, but also on the degree of substitution between banned and non-banned places. A recent study for Brazil finds that bans reduced smoking exposure for pregnant women working in the hospitality sector, leading to improved birth outcomes (Da Mata and Drugowick, 2023). This is compelling evidence of reduced passive smoking in targeted environments. Our findings on cessation (and reduced initiation) complement this prior literature, to the extent that they mitigate concerns about the potential displacement of smoking from public to private spaces. Taken together, these results strengthen the case for introducing comprehensive smoking bans to reduce both active and passive smoking.

The remainder of our paper is organized as follows. In Section 2, we first present background information about the smoking bans in Brazil. Subsequently, we discuss additional tobacco control policies and potential threats to identification. We provide supporting evidence

[3] Most prior studies rely on estimators that assume homogeneous effects across time and cohorts; however, our findings reveal that the effects of the bans differ along both dimensions. We also emphasize the importance of carefully examining trends that may be driving the implementation of smoking bans.

[4] Although some evidence suggests that cigarette taxes experienced as a teenager may reduce smoking in adulthood (Friedson et al., 2023), the current consensus is that both adults and youths are relatively unresponsive to price changes (DeCicca et al., 2022). Youths' price sensitivity may be even lower in LMICs, compared to high-income countries (Kostova, 2013). Further, alongside the introduction of minimum price laws and tax increases in Brazil, the share of illicit cigarette sales rose from less than 17% in 2008 to over 40% in 2019 (Iglesias et al., 2017; Divino et al., 2024).

[5] Refer to Section 3 for details about the data, Section 5.1 for methodological explanations, and Online Appendix A.3 for data validation.

[6] This finding aligns with research by Meier et al. (2021), which illustrates how teenagers often circumvent sales bans. In contrast, Pfeifer et al. (2020) found that smoking bans targeting schools are effective in preventing smoking initiation in Germany, with effects ranging from 14% to 22%.

[7] The original sample of PNS 2013 also includes individuals aged 30 and above, allowing us to estimate the bans' effects for this group. In unreported results, out of the scope of this paper, we find weaker effects for this age group: 3% increase in cessation; and a modest, albeit imprecise, reduction in prevalence. These findings are available upon request.

that these national-level policies do not confound our estimates of the effects of smoking bans. Our data is comprehensively described in Section 3. Section 4 introduces our empirical strategy and identifying assumptions. The results are presented and discussed in Section 5, and we conclude in Section 6.

2. Smoking bans in Brazil

Brazil has been implementing policies to reduce cigarette consumption since the 1980s through the establishment of the National Tobacco Control Program (INCA, 2023a). Smoke-free zones were introduced by a national law in 1996. However, this law, along with its accompanying regulations, permitted smoking areas or lounges and lacked enforcement mechanisms. According to INCA (2010), the 1996 regulation did not align with the World Health Organization (WHO) recommendations and was poorly adopted by the entertainment and hospitality industries. While the regulation required smoking areas to have proper ventilation and be adequately insulated from non-smoking common areas, evidence indicates that compliance with these provisions was insufficient (INCA, 2010).

As a consequence, in practice, smoke-free areas were not enforced in Brazil until 2009, when several states and municipalities implemented comprehensive local smoking bans. These policies differed from the national regulation in two key aspects. First, they entirely prohibited smoking in all enclosed or partially open public spaces, eliminating designated smoking lounges. Second, they introduced monitoring agencies and penalties for violations, thereby institutionalizing enforcement mechanisms.[8]

Following the trend of the local policies, a national smoking ban was approved in 2011, replacing the 1996 legislation across the country. However, this ban only took effect in December 2014, following regulation by the federal executive branch. In this paper, we leverage the staggered implementation of comprehensive smoking bans across Brazilian localities between 2009 and 2013 to evaluate the policy's impact on smoking behavior. Our analysis focuses on the period up to 2013, as no untreated units remain for comparison after 2014.

The unit of analysis in this study is the state capital cities. A city is considered "treated" if it has implemented a local smoking ban or is covered by a state-wide ban. Other municipalities are not analyzed because they are not individually identified in our dataset, which is representative only at the state capital level. Additionally, even if a state and its capital did not enact a smoking ban, we cannot rule out the possibility that other cities within the state implemented local bans.[9] Focusing on state capitals makes the analyses more tractable since their institutional settings are often better documented and observable. Moreover, state capitals concentrate a significant proportion of Brazil's urban population, and most of the establishments targeted by smoking bans. These cities are also hubs for formal employment, public services, and cultural and leisure activities, making them particularly relevant for this study.

2.1. Enforcement

The comprehensive local smoking bans share not only a common scope but also similar enforcement mechanisms. Monitoring is primarily conducted by municipal health agencies, which operate as subsidiaries of the Brazilian Health Regulatory Agency (ANVISA), and the state Consumer Protection Agency (PROCON). For state-level bans, state and municipal agencies are expected to collaborate. Violations can be reported anonymously to monitoring agencies or the police. Upon receiving a report, the monitoring agency inspects the establishment and may impose penalties for non-compliance. These penalties, directed at the person responsible for the establishment (e.g., the owner), range from warnings to fines, and in cases of repeated violations, may lead to the closure of the establishment.

However, effective enforcement depends on several factors: resource allocation for surveillance, public awareness and understanding of the law, and political will to penalize violators. The mere existence of a law does not guarantee its enforcement, particularly in developing countries where institutions are often weaker (Acemoglu et al., 2002). To evaluate the enforcement of local smoking bans in Brazil, we analyzed data from monitoring agencies across the country, including information on violations and penalties. Based on this data, we created a two-level enforcement index. The higher level represents bans with extensive dissemination, active surveillance, frequent notifications, and the application of fines. The lower level represents bans with limited dissemination, some surveillance activities, and no penalties.[10]

Fig. 1 presents the distribution of the local smoking bans across the Brazilian territory, including information about their year of implementation and level of enforcement. The markers show the location of each state capital and represent whether the local ban was introduced by the state, the capital, or both. The assignment of treatment status depends on the dates on which the local smoking ban became effective. For capitals with both state and capital-level bans, treatment assignment was based on the earliest effective date.[11] Once a smoking ban is adopted, it remains in effect for the entire study period. It is important to note that the majority of bans were implemented in 2009 (twelve out of fourteen), and half of these were strongly enforced. Eleven state capitals and the Federal District were not treated until 2014, when the national ban came into force. Further details are provided in Online Appendices A.1 and A.2.

2.2. Other tobacco control policies

Brazil is currently one of two countries in the world to have adopted all the tobacco control measures proposed by the WHO at the highest level of achievement (WHO, 2021). Besides smoke-free environments,

[8] Information on restrictions, effective dates, and enforcement rules was gathered directly from the laws. An initial list of local smoking bans was obtained from the non-governmental organization Tobacco Control Alliance (ACTBr, 2016). We also verified any changes to these laws and searched for unlisted ones through an internet review. Our findings align with those reported by Furtado and da Silva Filho (2016).

[9] The stratified sample distinguishes only the state capital from other municipalities within the state. No other cities are individually identified, and we do not observe whether or when smoking bans were enacted in all 5,570 Brazilian municipalities, as this information is not centralized.

[10] We gathered enforcement data from agencies of two states and seven state capitals. We complemented with information from newspaper reports for the remaining localities. The state of Rondônia (RO) approved a smoking ban in 2008 but never implemented it. The monitoring agency confirmed that the ban did not become effective until 2014 when the national ban took effect. As a result, it is excluded from our main analysis, although our results remain robust to its inclusion in the comparison group (Online Appendix Table B.11). A previous version of this paper classified Maceió/AL as part of the late-treated group (2011 cohort). However, since the current consensus indicates that smoking lounges were not prohibited until 2014, Maceió/AL is now considered untreated. This reclassification does not affect our estimates.

[11] For example, for the ten capitals where smoking bans took effect at the end of 2009, we assigned 2010 as the first year after treatment and 2009 as the baseline year. Two additional capitals, which implemented smoking bans between February and March of 2010, are also included in the main analysis with 2009 as the baseline. However, our results remain robust when these capitals are included in the 2010 cohort under a staggered treatment approach, alongside other late-treated capitals (Online Appendix Tables B.3 and B.4), or when they are excluded from the earlier-treated group (Table B.11).

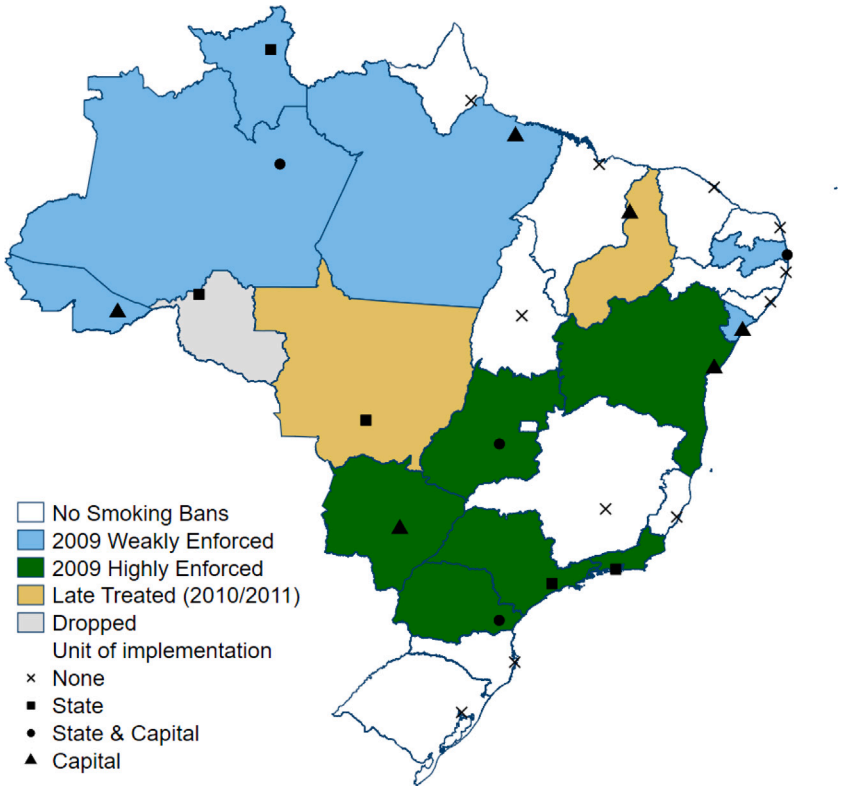

Fig. 1. Local smoking bans by year, unit of implementation, and enforcement level.
Notes: The colors represent the effective year and the level of enforcement of local smoking bans. For ease of visualization, the entire state is filled, even if the law was adopted at the capital level. The markers show the geographical location of the state capitals, and whether the smoking ban was adopted by the state, by the capital, or both. "Highly Enforced" bans were strongly disseminated and applied penalties. "Weakly Enforced" bans had some surveillance but no penalties. Rondônia was dropped from our main analysis since the ban approved in 2008 was never implemented. Data on local smoking bans come from ACTBr (2016), municipal and state legislation documents, and internet searches on public records and news. Enforcement levels were obtained from the monitoring agencies. Detailed information is provided in Online Appendix Tables A.1 and A.2.

they include (i) monitoring, (ii) warning (labels and mass media campaigns), (iii) advertising bans, (iv) cessation programs, and (v) taxation. In what follows, we provide supporting evidence that these policies are not a threat to the identification of the effects of the local smoking bans.

While warning labels and advertising restrictions were introduced by the 1996 national legislation, more comprehensive measures were only adopted in 2011. That year, cigarette advertising was completely banned in Brazil, and warning labels were required to cover a larger portion of the front of cigarette packs (INCA, 2023a). Since these are national policies and we are not aware of any local mass-media campaigns, warnings and advertising should not affect the municipalities differently. Further, as shown in Table 1, individuals living in capitals with local smoking bans did not report being more exposed to advertising or warnings in the 2013 survey, compared to those in capitals without smoking bans.

Cessation programs have been available since 2004 under the scope of the Ministry of Health, with their implementation assigned to municipalities (INCA, 2023b). We do not have access to data on the number of individuals who received treatment in each locality. However, around 11% of former smokers (4% among young adults) in our sample reported having attended a smoking cessation program (see Table 1). Importantly, there is evidence suggesting that participation in cessation programs was not higher in the treated capitals. In fact, if anything, lower participation in these programs would likely underestimate the effects of the smoking bans.

Finally, cigarette taxes and prices are also regulated at the national level. A minimum price has been in effect in Brazil since 2012, initially set at R$ 3.00 and increasing by R$ 0.50 each year up to R$ 5.00 (less than 1 USD) by 2016 (Receita Federal, 2016; INCA, 2023a). Comparing log prices reported by individuals in our survey, we find no evidence that prices were higher in the treated capitals (see Table 1). On the contrary, although imprecise, the price difference is negative among young adults. Additionally, in Online Appendix Figure B.1, we show that the distribution of prices in treated capitals is slightly more concentrated around lower prices compared to capitals without smoking bans. In the Online Appendix Figure B.2, we also demonstrate that the evolution of the cigarette price index from 2006 to 2013 was very similar between capitals with and without local smoking bans as of 2013.

Since any reported differences in the prevalence of warnings, cessation programs, and prices would be in contrast to the adoption of smoking bans, they should not be a concern for the assessment of the effects of local smoking bans in Brazil. If something, these differences would lead to an underestimation of the bans' impacts. These patterns remain similar when comparing only the capitals treated in 2009 with untreated capitals, and accounting for enforcement levels (Online Appendix Table B.1). We expand on this when discussing our results in Section 5.

3. Data

One of the challenges when studying smoking policies in Brazil is the absence of individual-level panel data on cigarette consumption. Without tracking individuals over time, for example, it is often not feasible to analyze initiation or cessation decisions. To overcome this limitation, we have created a panel with annual individual smoking status using retrospective information from the 2013 National Health Survey in Brazil (PNS) (IBGE, 2014), which contains smoking information from a representative sample of the population of each Brazilian state and their respective capitals. Retrospective data are widely used to study smoking behavior and provide an accurate measure of smoking status, especially among young individuals (Kenkel et al., 2003;

Table 1
Prevalence of other tobacco control policies in 2013.

2013 Survey	Adults			Young adults		
	Untreated	Diff. Treated	N	Untreated	Diff. Treated	N
Advertising	0.304	0.011	23,115	0.313	0.015	8,811
	(0.019)	(0.027)	23,115	(0.019)	(0.034)	
	.	[0.718]		.	[0.709]	
Warnings						
News Papers	0.330	−0.044	23,115	0.322	−0.052	8,811
	(0.034)	(0.039)	23,115	(0.036)	(0.039)	
	.	[0.347]		.	[0.254]	
TV	0.449	−0.035	23,115	0.434	−0.038	8,811
	(0.020)	(0.026)	23,115	(0.020)	(0.025)	
	.	[0.265]		.	[0.187]	
Radio	0.186	−0.008	23,115	0.161	−0.013	8,811
	(0.021)	(0.027)	23,115	(0.020)	(0.031)	
	.	[0.809]		.	[0.696]	
Package	0.556	0.002	23,115	0.571	−0.015	8,811
	(0.012)	(0.034)	23,115	(0.017)	(0.041)	
	.	[0.955]		.	[0.759]	
Cessation Programs	0.107	−0.020	1,641	0.037	−0.024	610
	(0.024)	(0.029)	1,641	(0.014)	(0.014)	
	.	[0.516]		.	[0.151]	
Log Price	1.625	−0.029	2,897	1.721	−0.094*	968
	(0.013)	(0.029)	2,897	(0.024)	(0.045)	
	.	[0.320]		.	[0.059]	

Notes: Prevalence of smoking-related policies based on self-reported data from the 2013 PNS. "*Untreated*" refers to the percentage of individuals who reported being exposed to advertising, warnings, or participating in smoking cessation programs (or reported cigarette prices in log) in capitals that did not have a local smoking ban at the time. "*Diff. Treated*" shows the differences in exposure to these policies between capitals that had a local smoking ban and those that did not. "*N*" Represents the number of survey respondents who answered the related questions. "*Adults*" are defined as individuals aged 18–65 at the time of the survey, while "*Young Adults*" are those aged 15–29 during the period from 2009 to 2013, as per the analysis focus. Prices are reported for a pack of cigarettes in the last purchase. All regressions use PNS sampling weights and report cluster-robust standard errors (at the capital level) in parentheses and the wild-cluster bootstrap p-values in square brackets. The treatment group consists of capitals that implemented the policy in 2009, along with two capitals treated later. Results comparing only the capitals treated in 2009 with untreated capitals, and accounting for enforcement levels, are presented in Online Appendix Table B.1.

Christopoulou and Lillard, 2015).

PNS identifies whether each individual was a regular smoker (smoking on a daily basis) at the time of the survey. In case an individual reports not being a regular smoker, then they need to respond if they have ever been one. For both cases, we observe the age at which individuals started smoking (i.e., their initiation age). For former regular smokers, we also observe the number of years/months since they quit smoking (i.e., allowing us to calculate cessation time). Note that we identify individuals who have completely quit smoking, rather than those who may have only reduced their smoking frequency. Some individuals may indicate that they are not regular smokers but smoke occasionally. These cases, which account for less than 2% of the 2013 sample, are excluded from our variables of initiation, cessation, and prevalence, which are defined strictly in terms of regular, daily smoking. Our variable specifically captures "daily smoking initiation", which does not allow us to distinguish between individuals who became regular smokers right away, versus those who were previously casual smokers but increased their smoking frequency to daily.

Using retrospective data from PNS, we constructed an individual-level panel tracking *regular smoking status* from 2005 to 2013. A detailed explanation is provided in the Online Appendix A, where Figure A.1 illustrates the survey's retrospective structure. In the Online Appendix A.3, we compare the variables obtained for 2008 through our approach (i.e., a "lagged" PNS) to those from the 2008 Tobacco Special Survey included in the National Household Survey (IBGE, 2009b). Overall, the results show that our lagged PNS closely matches the 2008 real-time survey, indicating that the constructed panel is reliable for the objectives of this study. The primary limitation of this approach is the absence of time-varying covariates at the individual level. Nevertheless, we argue that individual-specific effects capture the key sources of heterogeneity in this context. In addition, our results are robust to controlling for age squared and for household income per capita.

As explained in Section 2, our sample is restricted to individuals living in Brazilian state capitals. This sample constitutes around 45% of the individuals in PNS and represents 25% of the Brazilian population (more than 36 million adults in 2013). Given the focus of this paper, we have also restricted our analysis to young adults aged 15 to 29 in each year of the panel. These are the age cutoffs that define "young adults" according to the Brazilian Youth Statute.[12] Our final sample constitutes an unbalanced panel data with 11,308 individuals born from 1975 to 1995 and a total of 72,005 observations over the period 2005–2013.[13] Our main findings are robust to considering instead a balanced panel of individuals born from 1975 to 1995 (Online Appendix Table B.12).

3.1. Smoking behavior in Brazil

Brazil is a continental country, with the seventh largest population in the world and, consequently, a substantial number of smokers: more than 18.4 million regular smokers in 2013, of which almost 3.4 million were young adults. In addition, Brazilian overall smoking prevalence (14.7% in 2013) is similar to that of many developing countries, such as Mexico and other countries in Africa (WHO, 2016, 2019).

Due to the singular achievement of Brazilian tobacco control policies, the country is considered a success case by the WHO. As of

[12] National Law n. 1285/2013, which establishes the rights of young people, as well as the principles and guidelines for public policies targeting young adults. Even though the PNS module for smoking behavior was only applied to adults (18 years or older), our retrospective data procedure allows us to assess behavior at earlier ages.

[13] 33.7% of the individuals are observed in all the nine years of the panel; more than 50% are observed for seven periods or more; and around 25% are observed up to 4 periods, with only 5% in one period (those with 29 years old in 2005).

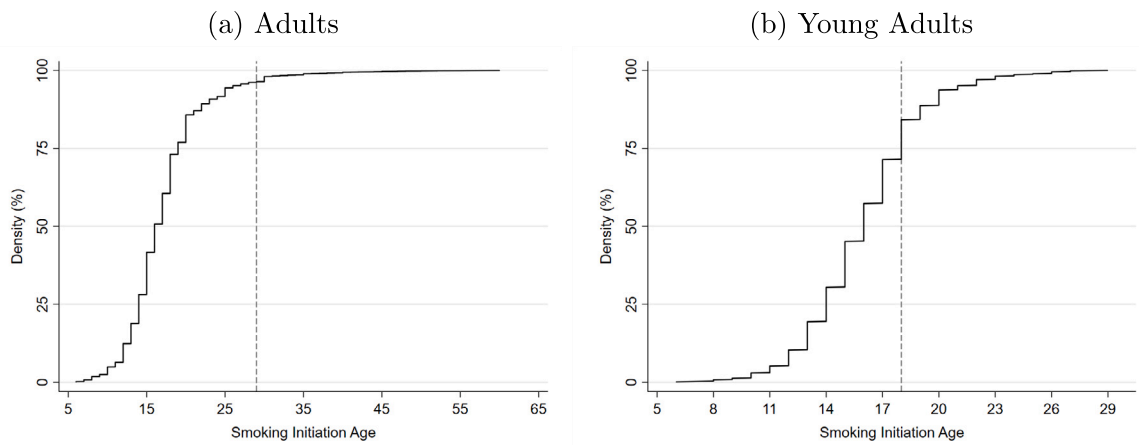

Fig. 2. Cumulative distribution of smoking initiation age.
Notes: Smoking initiation age for current and former regular smokers in 2013. Panel (a) includes adults aged 18 to 65 residing in Brazilian capitals, while Panel (b) focuses on young individuals aged up to 29 in these areas. Vertical dashed lines indicate that over 96% of individuals begin smoking by age 29 (Panel a), and nearly 72% of young adults started smoking daily before the legal age of 18 (Panel b).

2020, it was among the only two countries in the world to have fully implemented all the recommended measures (WHO, 2021). In addition, smoking prevalence has been following an expressive downward trend in the last decades in Brazil. For instance, nearly 15.6% of adults in the Brazilian states' capitals were regular smokers in 2005. This share decreased by 4.4 percentage points, reaching 11.2% by 2013. The trend is similar among young adults: from 12.5% regularly smoking in 2005 to 8.9% in 2013.

Our sample represents over 1.3 million young adults who were regular smokers in Brazilian capitals in 2009. By 2013, more than 20% of these individuals had quit smoking, while an additional 2% had started daily smoking between 2009 and 2013. Fig. 2 reveals a key insight into smoking behavior: initiation occurs essentially during early life. Among adults who were current or former regular smokers in 2013 (Panel a), 28% began smoking before the age of 15, and over 96% had started by age 29. Consequently, almost 70% of smoking initiation took place within the 15–29 age range, which is the focus of this study. When examining young adults specifically (Panel b), we find that nearly 72% started smoking daily before turning 18, which is the minimum legal age to purchase cigarettes and alcohol in Brazil.

This pattern motivates the focus on young adults when evaluating policies aimed at reducing smoking addiction. Since smoking behavior typically begins in early life, age is closely linked to the accumulated addictive stock. At early ages, since the addictive stock is relatively small, we should expect cessation to increase as a response to the costs imposed by the smoking bans. However, once addiction is settled, the present orientation of young adults may hinder cessation efforts, highlighting the importance of targeting initiation as a strategy to prevent long-term addiction. In the following sections, we analyze to what extent smoking bans are effective across those dimensions.

4. Empirical strategy

We estimate the impacts of local smoking bans implemented in Brazil from 2009 to 2013 on young adults residing in the treated capitals using regular smoking status from 2005 to 2013. Fig. 1 presents the treatment cohorts and the location of state capitals in Brazil. Although our focus on state capitals is due to data availability, the large distance between them attenuates spillover concerns. Furthermore, since migration is costly in Brazil (Oliveira and Pereda, 2020), it is unlikely that either people or companies would migrate between state capitals as a response to local smoking bans.[14]

In the context of this paper, we allow the impacts of smoking bans to vary across cohorts and over exposure time. This follows from the fact that smoking bans might take some time to be fully assimilated due to adaptation to legislation and the consequent social acceptance and control. Another source of potential heterogeneity is the level of enforcement of the bans, discussed in Section 4.2.1.

In our first analysis, we consider all the fourteen treated capitals using a staggered difference-in-differences approach. The comparison group consists of twelve capitals untreated until the national smoking ban came into force in 2014, thus "never-treated" for the purposes of this paper (see Fig. 1). We provide supporting evidence that the parallel trends assumption holds for the early-treated cohort (i.e., twelve capitals), but smoking followed different trends in the two late-treated capitals. Therefore, in our main analysis, we compare the twelve "early-treated" capitals with the "never-treated" group using a generalized difference-in-differences approach.

4.1. Staggered difference-in-differences

Define l as the time relative to the baseline year; e as the cohort related to each baseline year ($e = 2009, 2010, 2011$); and $SB_{i,e}$ as a dummy variable that equals 1 if individual i was residing in the capital in treatment cohort e at the moment of the survey, and 0 otherwise. Following Sun and Abraham (2021), we employ an event-study specification that accounts for heterogeneous effects across cohorts and over time. Specifically, the specification is an extended linear two-way fixed effects interacting period and cohort indicators, as follows[15]:

$$y_{i,t} = \mu_i + d_t + \sum_{e\in\{2009,2010,2011\}} \sum_{l\neq 0} \beta_{e,l}(\mathbb{1}\{l = t - e\} \cdot SB_{i,e}) + \epsilon_{i,t} \quad , \tag{1}$$

where μ_i are individual fixed effects, d_t are year fixed effects, and $\epsilon_{i,t}$ is an individual random error term. The dependent variable $y_{i,t}$ is a binary variable equal to 1 if individual i is a regular smoker in year t, and equal to zero otherwise.

[14] Based on a random sample of 3,456,597 individuals (615,544 young adults) from 2010 Census (IBGE, 2010), we observe that migration is more common within rather than between states. While 38% of young adults were living in a different municipality than where they were born, only 15% migrated between states, and even less, 7.8%, between treated and untreated states.

[15] In settings without additional covariates, this specification is equivalent to the proposal from Callaway and Sant'Anna (2021).

Within this framework, $\hat{\beta}_{e,l}$ is a consistent estimator for the average treatment effect of smoking bans on smoking prevalence for the treated cohort e after l years exposed to the bans, under two assumptions: (i) parallel trends, and (ii) no anticipatory effects. The parallel trends assumption implies that, in the absence of treatment, the outcomes for treated and never-treated groups would have followed the same trajectory. The no anticipatory effects assumption means that there are no changes in behavior before the bans are implemented (i.e., treatment effects are zero before treatment).

We find evidence supporting these assumptions for the early-treated cohort (i.e., twelve capitals treated from 2009 onward). First, state-level data from a 2008 survey indicate that smoking prevalence was similar between states with treated and untreated capitals before the implementation of local smoking bans. On average, 10.7% of young adults were smoking in 2008, with a (non-statistically significant) smaller prevalence by 1.1 percentage points in the treated group (Online Appendix Table B.2).

Since the identifying assumption relies on trajectories rather than levels, we also tested the joint significance of pre-treatment coefficients on smoking prevalence from Eq. (1) using an F-test (Wooldridge, 2021). For the 2009 cohort, we cannot reject the null hypothesis that the pre-treatment coefficients are jointly zero (see column (1) in Online Appendix Table B.3). However, for the 2010 and 2011 cohorts, the null hypothesis is rejected, even after controlling for cohort-specific linear trends (see column (2) in Online Appendix Table B.3). We interpret this as evidence of potential violations of the parallel trends assumption for the 2010 and 2011 cohorts, which is plausible given that these cohorts each consist of a single unit. Therefore, we exclude the late-treated capitals from our main analysis, but our results remain robust when including them under a staggered treatment design (Online Appendix Table B.4).

4.2. Generalized difference-in-differences

The main specification in this paper is based on a generalized difference-in-differences design, comparing the 2009 cohort with never-treated capitals, as follows:

$$y_{i,t} = \mu_i + d_t + \sum_{\substack{s=2006 \\ (s\neq 2009)}}^{2013} \beta_s(\mathbb{1}\{s=t\} \cdot SB_i) + \delta(SB_i \cdot Trend) + \epsilon_{i,t} \quad , \tag{2}$$

where SB_i is a dummy variable that equals 1 if individual i is exposed to a smoking ban from 2009 onward, and zero if individual i was not exposed to a local smoking ban during the period of study. Even though we are not able to reject the assumption that pre-treatment parameters are jointly or individually null, we are aware that this might be due to the low power associated with pre-trends testing (Roth, 2022). We address potential differences in pre-trends by controlling for treated-cohort specific linear trend, which is captured by the interaction of the treatment dummy (SB_i) with $Trend$ (Wooldridge, 2021). Our results support the linear specification of pre-trends, and we discuss its potential drivers in the next section. We show that differential trends are driven by increasing initiation trajectories among early adopters, and our results are consistent when removing them from the sample. Our estimates are also consistent to controlling for age square as a time-varying covariate.

Eq. (2) assesses the effects of smoking bans on smoking prevalence among young adults residing in the capitals treated by a local ban. The coefficient β_t captures post-treatment effects for the years 2010 to 2013, and pre-treatment deviations from linear trends for 2006 to 2008. The interaction for $t = 2005$ is additionally omitted when accounting for linear trends. For initiation and cessation, we use the same equation and outcome but apply additional sample restrictions. For example, for initiation, we restrict the sample to non-smokers at baseline (i.e., 2009). Conversely, for cessation, we restrict the sample to regular smokers in 2009.

We applied PNS weighting adjustments outlined by the survey designers to ensure that our analysis is both representative and unbiased.[16] These weights account for selection probabilities, non-response, and post-stratification adjustments based on key demographic variables, correcting for potential biases related to respondent replacements (Souza-Júnior et al., 2015). We additionally performed sensitivity analyses to address potential confounding factors, such as socioeconomic status, which could influence smoking behavior and survey response rates (see Online Appendix Table B.13). These analyses confirm that the observed effects are indeed attributable to smoking bans and enforcement policies, rather than sample composition or response biases.

Finally, since the policy is implemented at the state or capital levels, we allow the errors to be correlated among individuals who reside in the same capital by estimating cluster-robust standard errors. Due to concerns about the small number of clusters, we also report p-values from Wild-Cluster Bootstrap (Cameron and Miller, 2015; MacKinnon and Webb, 2017) using the `boottest` command built for Stata by Roodman et al. (2019).

4.2.1. Heterogeneity by enforcement level

We also study the extent to which the effects of smoking bans depend on their level of enforcement. Among the 2009 cohort, six capitals are treated by strongly enforced bans, while six are treated with low enforcement, as shown in Fig. 1. We interact treatment indicators for each enforcement level (i.e., low and high) with year indicators, as follows:

$$y_{i,t} = \mu_i + d_t + \sum_{\substack{s=2006 \\ (s\neq 2009)}}^{2013} \sum_{v\in\{low,high\}} \beta_{s,v} \cdot \mathbb{1}\{s=t\} \cdot SB_{i,v} + \sum_{v\in\{low,high\}} \delta_v \cdot SB_{i,v} \cdot Trend + \epsilon_{i,t} \tag{3}$$

where $SB_{i,v}$ is equal 1 if individual i was residing in a capital treated with a local smoking ban enforced with level $v = \{low, high\}$, and zero otherwise. Note that we interact the linear trend variable (*Trend*) with treatment dummies specific for each enforcement level. Therefore, we allow the treatment groups with different enforcement levels to follow specific linear trends.

5. Results

We first estimate the effects of smoking bans on smoking prevalence among young adults in Brazilian capitals from 2005 to 2013. The results shown in Fig. 3(a) indicate that the local smoking bans are associated with a break in their smoking trajectory, manifesting after three years of exposure to the bans. Our lower-bound estimates reveal a 0.9 percentage point reduction in smoking prevalence by 2012 and a 1.1 percentage point reduction by 2013. The latter represents a 9% decrease in the average smoking prevalence among young adults in the treated cohort, which was 12% in 2009. When accounting for the increasing trend in smoking prevalence, the effects become more pronounced, reaching up to 2.2 percentage points by 2013. This suggests that smoking bans may have reduced smoking prevalence by up to 18% among young adults in the treated cohort after four years of exposure. The fact that the impacts were not immediate is consistent with a scenario in which short-run impacts may not be significant due to delays in enforcement and due to the addictive nature of cigarette consumption and its consequent slow adjustment. The effects in 2012 and 2013 are also statistically significant according

[16] Sampling weights are essential for obtaining aggregate smoking prevalence statistics consistent with the target Brazilian sub-population. Without these weights, smoking prevalence among young adults would be underestimated by nearly 1 percentage point, and the treated group would be underrepresented.

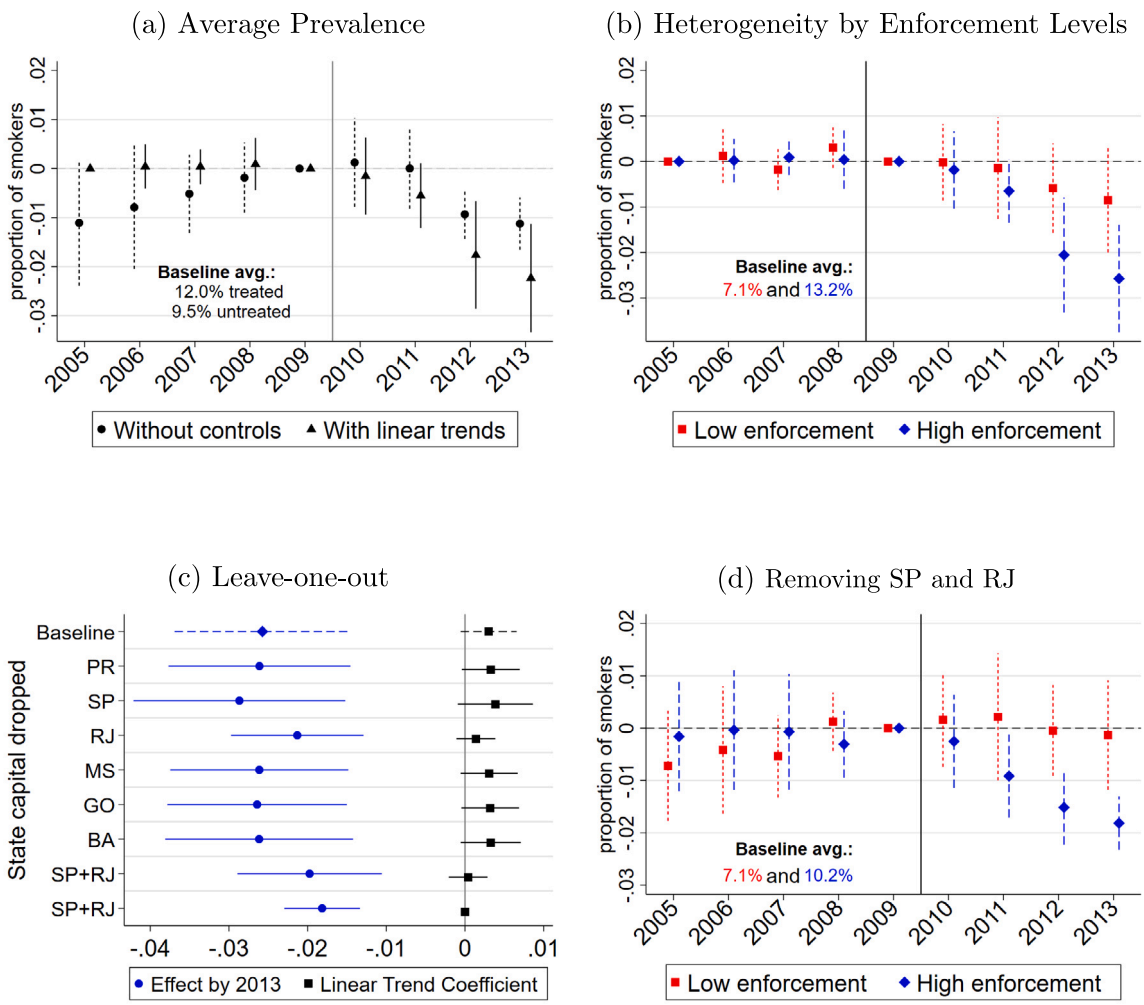

Fig. 3. Impacts of smoking bans on smoking prevalence.
Notes: **Panel (a)**: estimates from Eq. (2). P-values from the wild-cluster bootstrap are provided in Online Appendix Table B.5. **Panel (b)**: estimates from Eq. (3). The results excluding linear trends and p-values from the wild-cluster bootstrap are provided in Online Appendix Table B.6. **Panel (c):** reports only the 2013 coefficients for highly enforced bans. The remaining coefficients, standard errors, and wild-cluster bootstrap p-values can be found in Online Appendix Table B.7. **Panel (d):** estimates are shown without controlling for linear trends after excluding São Paulo (SP) and Rio de Janeiro (RJ). All regressions include year and individual fixed effects, use PNS sampling weights, and display 95% confidence intervals from cluster-robust standard errors at capital. The main sample includes 68,130 observations across 10,700 individuals.

to confidence intervals constructed using wild-cluster bootstrap. These effects are also mostly unchanged after controlling for the square of age (Appendix Table B.5).

Importantly, the average effects mask significant heterogeneity regarding the effective implementation of local smoking bans. Fig. 3(b) suggests that the prior results were predominantly driven by strongly enforced bans, which lead to a 20% reduction in smoking prevalence after four years of exposure. In contrast, the effects of weakly enforced smoking bans were considerably smaller and statistically indistinguishable from zero. These results highlight the importance of considering enforcement levels in policy evaluations in developing countries, where institutional capabilities are often weaker. The experience of Brazilian smoking bans demonstrates that public policies can be effective when properly implemented in such settings. The enforcement mechanisms from the successful capitals may provide valuable insights for other developing countries when implementing similar regulatory policies. Additionally, the success of the Brazilian smoking bans may be informative even for developed countries where such bans are narrower in scope (e.g., some countries in Europe). Our paper helps to disentangle the mechanisms through which comprehensive bans might have effectively transferred property rights over "clean air" from smokers to non-smokers in Brazil, recently even leading to discussions around smoking bans in open common areas (such as parks and public squares).[17]

A possible concern with the prior heterogeneity analysis is that strongly enforced bans are concentrated in capitals with larger smoking prevalence and higher population density. To address this, we implement a leave-one-out analysis, examining whether our results were driven by any particular capital. We re-estimate the heterogeneity specifications on several sub-samples, sequentially removing one capital with a strongly enforced smoking ban. The effects shown in

[17] Da Mata and Drugowick (2023) show that the bans reduced passive smoking in targeted environments. Further, the municipality of São Paulo banned smoking from public parks in 2019. Another example is the state of Santa Catarina, which banned smoking from parks and playgrounds in April 2024.

Fig. 3(c) remain consistent with our main findings. We still find that strongly enforced smoking bans reduced smoking prevalence by over 18%, even after simultaneously removing both São Paulo (SP) and Rio de Janeiro (RJ), the main cities in Brazil, and the leaders in adopting comprehensive smoking bans.

Although the baseline pre-treatment coefficients are not statistically different from zero, the trajectory of smoking prevalence from 2005 to 2009 suggests an increasing trend in the treated cohort. This motivates the inclusion of treated-cohort-specific linear trends in our main specifications.[18] In Fig. 3(c), the coefficient of the linear trend becomes practically null after removing SP and RJ, suggesting that it was driven by these leading regions.

While smoking bans may have been implemented in response to differential smoking trajectories in SP and RJ, Fig. 3(d) offers supporting evidence that adoption in the remaining regions aligns with the parallel trends assumption. After removing SP and RJ, the unconditional pre-treatment coefficients become nearly zero, especially for strongly enforced bans, and there is a decreasing trajectory in smoking prevalence after the treatment, leading to a 15% reduction by 2012 and an 18% reduction by 2013. This finding supports the causal interpretation of our estimates, demonstrating that they are not entirely driven by the main cities, nor dependent on a linear extrapolation of pre-trends.

In what follows, we decompose the effects on smoking prevalence between effects on initiation and cessation and study the response to the bans across these margins of adjustment.

5.1. Decomposition of the effects on smoking prevalence

The change in smoking prevalence from 2009 to year t can be expressed by the difference in the number of smokers normalized by the population (N):

$$\Delta prevalence_t \equiv \frac{\sum_i \mathbb{I}[i \text{ smokes in } t]}{N} - \frac{\sum_i \mathbb{I}[i \text{ smokes in } 2009]}{N} \tag{4}$$

Since each individual that smoked in year t was either smoking or not in 2009, we can rewrite the first numerator as follows:

$$\sum_i \mathbb{I}[i \text{ smokes in } t] = \sum_i \mathbb{I}[i \text{ smokes in } t] \times (\mathbb{I}[i \text{ smokes in } 2009] + \mathbb{I}[i \text{ does not smoke in } 2009])$$

Likewise, the number of smokers in 2009 can be expressed in terms of individuals continuously smoking in t and those that had quit by t:

$$\sum_i \mathbb{I}[i \text{ smokes in } 2009] = \sum_i \mathbb{I}[i \text{ smokes in } 2009] \times (\mathbb{I}[i \text{ smokes in } t] + \mathbb{I}[i \text{ does not smoke in } t])$$

Once we take the difference between these terms, the number of individuals continuously smoking cancels out. Therefore, we can express the numerator in (4) as the difference between the number of individuals smoking in t that were not smoking in 2009, and the number of individuals that were smoking in 2009, but not in t. Normalizing both terms by the number of non-smokers (NS) and smokers (S) in 2009, respectively, we get:

$$\sum_i \mathbb{I}[i \text{ smokes in } t] - \sum_i \mathbb{I}[i \text{ smokes in } 2009] = \underbrace{\frac{\sum_i \mathbb{I}[i \text{ smokes in } t, \text{ does not smoke in } 2009]}{NS}}_{\text{Initiation}} \cdot NS - \underbrace{\frac{\sum_i \mathbb{I}[i \text{ smokes in } 2009, \text{ does not smoke in } t]}{S}}_{\text{Cessation}} \cdot S \tag{5}$$

Eq. (5) expresses the variation in the number of smokers in terms of initiation and cessation. *"Initiation"* is given by the share of individuals in the sample of non-smokers in 2009 who are smokers in t. *"Cessation"* is the share of non-smokers in t in the sample of individuals who were smoking in 2009. The effect on smoking prevalence is equivalent to the effect on initiation normalized by the share of non-smokers in 2009 (NS/N), *minus* the effect on cessation normalized by the share of smokers in 2009 (S/N):

$$\Delta prevalence_t = Initiation \cdot \frac{NS}{N} - Cessation \cdot \frac{S}{N} \quad , \tag{6}$$

where $S + NS = N$.

Based on this decomposition, we estimate the effects on *smoking cessation* by restricting the sample to young adults who were smoking in 2009, such that the outcome is equal to one for all selected individuals in 2009. In subsequent years ($t = 2010, \ldots, 2013$), the outcome is equal to one if the individual is still smoking and zero otherwise. To reflect cessation patterns, we multiply the coefficients by (−1).[19] An important drawback is that, by restricting the sample to smokers in 2009, the trajectories in the pre-treatment years will not reflect cessation patterns. Likewise, pre-trends in the sub-sample of non-smokers in 2009 will not capture the trajectories in *smoking initiation* before the treatment. The next subsection explains how we study these pre-trends using placebo samples.

5.1.1. Placebo estimates for pre-treatment trajectories

We compute placebo estimates for the pre-treatment trajectories in cessation and initiation by restricting the sample to smokers and non-smokers in 2005, respectively. We then regress the outcome of these individuals on a treatment indicator interacted with year dummies and with a linear trend variable using data up to 2009.[20] The results are shown in Table 2.

Based on the sample of non-smokers in 2005, we find supporting evidence that initiation was increasing more in the treated cohort before the introduction of the local smoking bans. Even though imprecisely, this can be observed by the point estimates without controlling for linear trends in column (1), and by the direction and magnitude of the linear trend coefficient in column (2). Remarkably, the null coefficients for the year dummies column (2) indicate that the linear trend is a good approximation to the pre-trend in initiation. In contrast, taking into account that the baseline smoking prevalence is 1.00 in the sample of smokers in 2005, the coefficients on cessation are relatively smaller in magnitude. This suggests that the unconditional parallel trends assumption is more likely to hold for smoking cessation than for initiation.

Altogether, these findings point to initiation as the main driver of the slower decrease in smoking prevalence observed in the treated

[18] The linear trend coefficients are estimated as the differences in prevalence growth from 2005 to 2009. There is supporting evidence that the differential pre-trends are fully captured by linear trends. After controlling for it, the remaining pre-treatment coefficients become null, suggesting no deviations from the linear specification (Online Appendix Tables B.5 and B.6). In addition, when fitting the linear trend using all the pre-treatment periods (i.e., equalizing all coefficients to zero), the estimates for the linear trend and the post-treatment coefficients are unchanged.

[19] We multiply the coefficient by (−1) because our dependent variable – regular smoke – reflects smoking prevalence rather than cessation. This approach is equivalent to defining an alternative cessation variable as an indicator that is 0 for smokers and switches to 1 upon quitting.

[20] For instance, following individuals for who $y_{i,2005} = 0$ from 2005 to 2009, we estimate for initiation: $y_{i,t} = \mu_i + d_t + \sum_{s=2006}^{2008} \beta_s^I \times SB_i \times \mathbb{I}[s = t] + \delta^I Trend \times SB_i + \epsilon_{i,t}$. See Online Appendix C for further discussion and formalization.

Table 2
Placebo estimates of smoking bans on cessation and initiation.

	Initiation (2005 non-smokers)			Cessation (2005 smokers)		
	(1)	(2)	(3)	(4)	(5)	(6)
2005	·	·	·	·	·	·
2006	0.004	0.001	0.002	−0.001	0.004	0.004
	(0.003)	(0.002)	(0.002)	(0.013)	(0.014)	(0.012)
	[0.179]	[0.584]	[0.294]	[0.978]	[0.780]	[0.771]
2007	0.006*	0.001	0.001	−0.008	0.002	0.009
	(0.004)	(0.001)	(0.002)	(0.012)	(0.012)	(0.012)
	[0.097]	[0.672]	[0.769]	[0.650]	[0.878]	[0.460]
2008	0.010*	0.002	0.001	−0.016	−0.001	0.013
	(0.005)	(0.002)	(0.002)	(0.018)	(0.015)	(0.014)
	[0.091]	[0.519]	[0.465]	[0.476]	[0.944]	[0.432]
2009	0.012	·	·	−0.019	·	·
	(0.007)	·	·	(0.024)	·	·
	[0.151]	·	·	[0.488]	·	·
Trends	·	0.003	0.001	·	−0.005	0.004
	·	(0.002)	(0.001)	·	(0.006)	(0.008)
	·	[0.151]	[0.623]	·	[0.488]	[0.626]
SP + RJ	Yes	Yes	No	Yes	Yes	No
F-stat	3.753	0.812	0.735	0.245	0.111	0.399
P-value	0.110	0.584	0.621	0.962	0.961	0.837
2009 mean	0.041		0.026	0.903		0.854

Notes: The *Initiation* sample is limited to non-smokers in 2005 (36,651 observations across 9,212 individuals), while the *Cessation* sample is restricted to smokers in 2005 (4,399 observations across 1,049 individuals). Estimates come from Eq. (2), using data from 2005 to 2009, with 2005 as the baseline. Columns (2)–(3) and (5)–(6) adjust for linear trends interacted with the treatment indicator (coefficient *Trends*). When controlling for linear trends, the 2005 and 2009 coefficients are normalized to zero. Columns (3) and (6) exclude São Paulo (SP) and Rio de Janeiro (RJ). All regressions include year and individual fixed effects, use PNS sampling weights, and report F-statistics and corresponding p-values for joint significance tests. Cluster-robust standard errors (at the capital level) are in parentheses, with wild-cluster bootstrap p-values in square brackets. Significance levels: * 10%; ** 5%; *** 1%.

capitals before the bans were introduced. However, when the leading adopters, SP and RJ, are removed from the sample, as shown in column (3), the linear trend slope gets smaller. This indicates that the differential trend is primarily driven by initiation patterns in these two cities, consistent with our findings for prevalence.

A possible explanation for greater initiation rates is that young adults may have been more engaged in risky behaviors in the treated capitals. We provide insight on this by analyzing responses to specific questions of the 2013 PNS which capture these behaviors (Table 3). For example, we find that the share of smokers reporting purchasing cigarettes below the minimum legal price, potentially from *smuggling*, was 13 percentage points higher in the group treated by strongly enforced bans, even though prices and taxes are set nationally (Section 2.2).[21] There is also evidence that a smaller share of smokers tried to quit in the treated capitals.

Since the above questions about risky behaviors were restricted to smokers in 2013, the observed differences could be partially related to selection. It is likely that young adults who were still smoking in 2013, despite the costs imposed by the bans, were also more prone to risky behaviors. However, results from the bottom panel of Table 3 suggest that selection might not explain all the differences. With a sample that includes both smokers and non-smokers, although imprecise, we find

Table 3
Relationship between smoking bans and other risky behaviors in 2013.

	Average	Difference for treated		
	Control	High enforced	Low enforced	N
Daily or causal smoker	0.112	0.006	−0.013	5,838
	(0.014)	(0.019)	(0.015)	
		[0.760]	[0.432]	
Smuggling	0.066	0.125***	0.119**	628
	(0.009)	(0.041)	(0.050)	
		[0.009]	[0.026]	
Tried to quit	0.506	−0.099**	−0.129	664
	(0.038)	(0.043)	(0.072)	
		[0.047]	[0.218]	
Initiation Age	15.977	0.296	1.056	540
	(0.246)	(0.262)	(0.688)	
		[0.344]	[0.254]	
Smokers + non-smokers				
Practice sport freq.	0.371	−0.040	−0.037	5,838
	(0.019)	(0.023)	(0.024)	
		[0.156]	[0.156]	
Consumes alcohol freq.	0.130	0.028	−0.023	5,838
	(0.014)	(0.019)	(0.025)	
		[0.167]	[0.496]	

Notes: Based on data from the 2013 PNS. "*Smuggling*" is measured as a binary variable, equal to 1 if the reported price for a pack of cigarettes is below the minimum legal price (3.50 Brazilian Reais), and 0 otherwise. Cluster-robust standard errors (at the capital level) are in parentheses, with wild-cluster bootstrap p-values in square brackets. Significance levels: * 10%; ** 5%; *** 1%.

that young adults who faced strongly enforced bans may be more likely to consume alcohol and less likely to practice sports regularly. Overall, these results could be explained by young adults in RJ and SP being generally more exposed to situations that can trigger smoking initiation (e.g., social interactions with alcohol consumption, availability of cigarettes at lower prices, etc.).

5.1.2. *Effects on smoking initiation and cessation*

By implementing the decomposition described in Eq. (6), we assess the effects of smoking bans on initiation by restricting the sample to non-smokers in 2009. The results from the previous section suggest that, in the absence of behavioral or environmental changes, a larger share of individuals would have become regular smokers in the capitals that implemented smoking bans. This would lead to the underestimation of the effects of the bans on initiation if not accounted for. We thus estimate the bans' causal effects as deviations from the treated cohort-specific linear trend.

However, because pre-treatment trajectories from the sub-sample of 2009 non-smokers do not capture trends in initiation, our standard event-study approach from Eq. (2) will not deliver consistent estimates. For instance, when restricting the sample to smokers in 2009 to study effects on cessation, pre-treatment trajectories would be partially capturing the positive trend in initiation. In our standard estimation, this trend would be abated from the post-treatment outcomes, leading to an overestimation of the effects on cessation. This raises the question of how to control for linear trends when outcomes are asymmetrical before and after the treatment.

Our approach consists of discounting, from the post-treatment outcomes, the linear trend predicted in the placebo samples.[22] Let $\hat{\delta}^I$ and $\hat{\delta}^C$ be the linear trend estimates for initiation and cessation from columns (2) and (5) of Table 2 respectively. For the heterogeneity by enforcement level, the linear trend coefficients are specific for strongly and weakly enforced groups (Online Appendix Table B.8).

[21] Access to cheaper, often illicit, cigarettes aligns with findings from Meier et al. (2021), which highlight how teenagers frequently circumvent sales bans. We do not control for prices, as data were only collected from smokers in 2013. However, evidence from a price index suggests that legal prices did not change differently among the treated and untreated groups over time (Online Appendix Figure B.2). Moreover, while youths may be more sensitive to price changes than adults (Carpenter and Cook, 2008; Friedson et al., 2023), the current consensus indicates that both adults and youths exhibit relatively low responsiveness to price changes (DeCicca et al., 2022).

[22] Further details are provided in Online Appendix C. The prediction of linear trends in a placebo pre-treatment sample and their imputation in the post-treatment data is also employed by Bhuller et al. (2013) and Goodman-Bacon (2021).

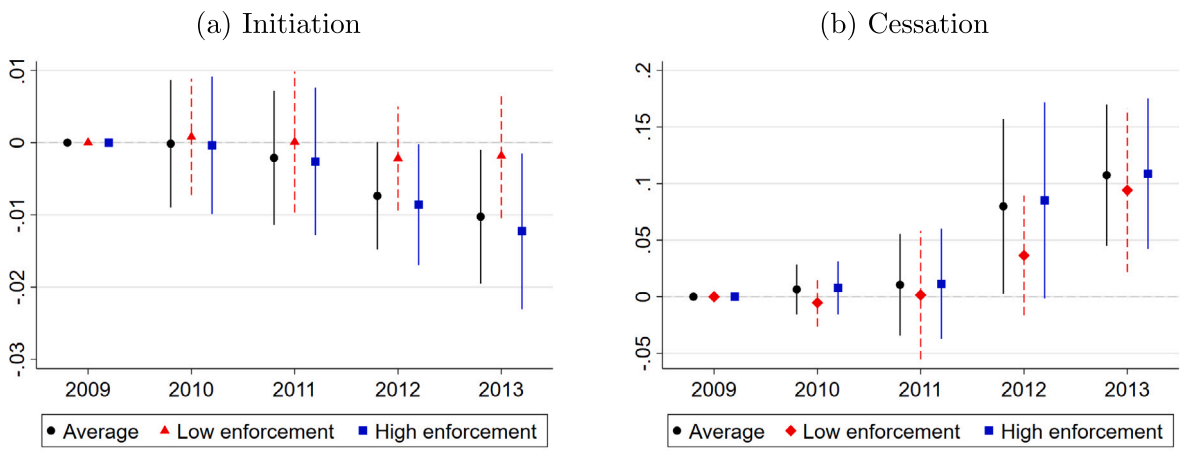

Fig. 4. Impact of smoking bans on smoking initiation and cessation.
Notes: All regressions include year and individual fixed effects, use PNS sampling weights, and display 95% confidence intervals from cluster-robust standard errors at capital. Estimates by enforcement level and wild-cluster bootstrap p-values can be found in Online Appendix Table B.8. Post-treatment coefficients are estimated using the sub-sample of non-smokers (panel a) and smokers (panel b) in 2009, with data from 2009 to 2013. Example equations for average estimates: $y_{i,t}^{\{I \text{ or } C\}} = \mu_i + d_t + \sum_{s=2010}^{2013} \beta_s \left(\mathbb{I}(s=t) \cdot SB_i\right) + \epsilon_{i,t}$. The outcomes are $y_{i,t}^{I} = y_{i,t} - \hat{\delta}^{I} Trend$ for non-smokers in 2009, and $y_{i,t}^{C} = y_{i,t} - \hat{\delta}^{C} Trend$ for smokers in 2009, where $\hat{\delta}^{I}$ and $\hat{\delta}^{C}$ are the *Trends* coefficients in columns (2) and (5) of Table 2 respectively.

For each individual i that was not smoking in 2009, we obtain the post-treatment outcome as follows:

$$y_{i,t}^{I} = y_{i,t} - \hat{\delta}^{I} Trend \times SB_i, \quad \text{for } t \geq 2009 \tag{7}$$

Likewise, the post-treatment outcome for each individual i that was smoking in 2009 is obtained as follows:

$$y_{i,t}^{C} = y_{i,t} - \hat{\delta}^{C} Trend \times SB_i, \quad \text{for } t \geq 2009 \tag{8}$$

We then regress $y_{i,t}^{I}$ and $y_{i,t}^{C}$ on individual and time fixed effects, and on time dummies interacted with the treatment indicator using data from 2009 to 2013. The results shown in Fig. 4(a) indicate a decreasing trend in initiation following the introduction of smoking bans, reaching a reduction of 1 percentage point by 2013. Importantly, this effect is primarily driven by strongly enforced bans, which reduced initiation by 1.2 percentage points by 2013 — equivalent to 25% of the 4% average initiation rate observed between 2005 and 2009. Furthermore, following young adults who were regular smokers in 2009, we find that smoking bans effectively promoted cessation. As shown in Fig. 4(b), the bans increased the likelihood of quitting. The impact of strongly enforced bans on smoking cessation is comparable to the average effect, resulting in nearly 11% of smokers quitting by 2013.

While weakly enforced bans were not effective in preventing initiation, Fig. 4(b) suggests that they may have increased cessation by over 9% by 2013. However, due to evidence against the parallel trends assumption for smoking initiation in the low-enforcement group, we refrain from making causal interpretations.[23] Nonetheless, as an exploratory exercise, assuming that the cessation effects are causal and initiation effects are null, this would imply a reduction in smoking prevalence by less than 10%. Besides being smaller, this estimate is more imprecise compared to the effects of strongly enforced bans. As a conclusion, smoking bans were more effective in shifting smoking prevalence when strongly enforced (if not only effective under this condition). The results for weakly enforced bans are inconclusive, but they might suggest that informal enforcement also has some role in the application of smoking bans.

Overall, our findings suggest that the introduction of smoking bans shifted the trajectory of smoking prevalence primarily through the promotion of smoking cessation. The decomposition analysis in Table 4 reveals that cessation accounts for nearly 60% of the total impact of the bans on smoking prevalence. The smaller effects on initiation may be partly because, as shown in Fig. 2, smoking initiation in Brazil often happens before the minimum legal age for drinking and smoking (i.e., 18 years old). The implication is that initiation might be happening at places that are not targeted by smoking bans.

Finally, the effects on prevalence derived from Eq. (6) and displayed in the third column of Table 4 closely mirror our baseline estimates in the fourth column. Additionally, inference remains robust when bootstrapping the entire procedure (p-values shown in curly brackets). This supports the validity of our approach to controlling for linear trends when outcomes are asymmetrical and offers valuable insights for researchers studying similar types of outcomes. We further discuss the implications of pre-trends in such settings in the Online Appendix C.

5.1.3. *Effects on cessation by addiction level*

Motivated by the theory of rational addiction (Becker and Murphy, 1988), we also examined how the effects on cessation vary with addiction levels, using the number of years an individual has been smoking regularly as a measure. Within this framework, whether smoking bans might affect cessation depends on the trade-off between two channels. On the one hand, current consumption depends on past consumption, which accumulates in the form of an "addictive stock" and *reinforces* addiction. On the other hand, depending on the extent to which individuals discount the future, this stock may also raise the marginal cost of consumption due to expected detrimental health effects. Because young individuals are more present-orientated, they might be more susceptible to addiction through the first channel. This suggests that the bans might be more effective in promoting cessation before addiction is settled (i.e., before the "addictive stock" is too large).

We estimate the effects on sub-samples of young adults who were smoking for a different length of time (i.e., with different levels of smoking addiction) when the bans were implemented. The results are shown in Table 5. The average effects discussed before are displayed in column (1) for comparison. In column (2), we look at cessation among young adults who started smoking just before the bans were implemented (i.e., at most three years of smoking by 2009). Columns (3) and (4) present the results for those who had been smoking for at least four years when the bans were implemented.

Our findings align with theoretical predictions, suggesting that smoking bans primarily promoted cessation among young adults who

[23] The p-value for joint significance of pre-treatment coefficients is equal to 0.076, even after controlling for linear trends (Online Appendix Table B.8). The rise in smoking initiation just before the bans may bias the estimated cessation effects if these individuals were more likely to quit shortly afterward.

Table 4
Decomposition of smoking prevalence.

			Prevalence	
	Initiation × 0.88 (1)	Cessation × 0.12 (2)	Decomposition (3)	Estimation (4)
2010	−0.001	0.001	−0.001	−0.002
	(0.004)	(0.001)	(0.004)	(0.004)
	{0.973}	{0.484}	{0.830}	[0.723]
2011	−0.002	0.001	−0.003	−0.006*
	(0.003)	(0.002)	(0.003)	(0.003)
	{0.553}	{0.590}	{0.282}	[0.093]
2012	−0.006**	0.010*	−0.016**	−0.018***
	(0.003)	(0.005)	(0.007)	(0.006)
	{0.043}	{0.078}	{0.025}	[0.004]
2013	−0.009**	0.013**	−0.022***	−0.022***
	(0.004)	(0.005)	(0.007)	(0.006)
	{0.032}	{0.016}	{0.003}	[0.003]
$N \times T$	31,523	3,447	68,130	
N	7,467	862	10,700	

Notes: The effects on initiation and cessation are estimated using sub-samples of non-smokers and smokers in 2009, respectively, with data from 2009 to 2013. In columns (1) and (2), the effects on initiation and cessation are weighted by the shares of non-smokers ($\frac{NS}{N} = 0.88$) and smokers ($\frac{S}{N} = 0.12$) in the treated group in 2009. The decomposition of the effect on prevalence in column (3) equals (1) *minus* (2). The effects on prevalence in column (4) are based on Eq. (2) using data from 2005 to 2013 and controlling for linear trends interacted with the treatment indicator. All regressions include year and individual fixed effects, use PNS sampling weights, and report cluster-robust standard errors (at the capital level) in parentheses. P-values from a bootstrap of the entire procedure (i.e., accounting for the first stage) are shown in curly brackets, while wild-cluster bootstrap p-values are in square brackets in column (4). Significance levels: * 10%; ** 5%; *** 1%.

Table 5
Impacts on smoking cessation by addiction level.

	(1) Full sample	(2) Addiction level	(3)	(4)
	2009 smokers	≤ 3 years	{4, 5} years	≥ 6 years
2010	0.006	0.024	−0.008	0.002
	(0.011)	(0.033)	(0.016)	(0.017)
	[0.574]	[0.548]	[0.789]	[0.931]
2011	0.011	0.103**	−0.070	−0.019
	(0.023)	(0.036)	(0.091)	(0.023)
	[0.672]	[0.017]	[0.655]	[0.478]
2012	0.080*	0.247***	−0.073	0.020
	(0.039)	(0.041)	(0.097)	(0.056)
	[0.084]	[0.001]	[0.524]	[0.671]
2013	0.107**	0.275***	−0.073	0.054
	(0.032)	(0.043)	(0.096)	(0.043)
	[0.024]	[0.001]	[0.509]	[0.271]
$N \times T$	3,447	884	485	2,078
N	862	184	101	577

Notes: The effects on cessation are estimated using sub-sample of smokers in 2009, with data from 2009 to 2013, based on the following equation: $y^C_{i,t} = \mu_i + d_t + \sum_{s=2010}^{2013} \beta_s \left(\mathbb{I}(s=t) \cdot SB_i\right) + \epsilon_{i,t}$. The outcome is $y^C_{i,t} = y_{i,t} - \hat{\delta}^C Trend$, where $\hat{\delta}^C$ is the *Trends* coefficient from column (5) of Table 2. All regressions include year and individual fixed effects, use PNS sampling weights, and report cluster-robust standard errors (at the capital level) in parentheses and the wild-cluster bootstrap p-values in square brackets. The outcomes consist of smoking prevalence among (1) young adults who were smoking in 2009; (2) young adults who had been smoking for up to 3 years in 2009; (3) young adults who had been smoking for 4 or 5 years in 2009; and (4) young adults who had been smoking for 6 or more years in 2009. The results remain robust when controlling for age squared (Online Appendix Table B.9) and when using a single regression that interacts treatment with addiction level (Online Appendix Table B.10). Significance levels: * 10%; ** 5%; *** 1%.

had been smoking for a relatively short period (i.e., at most three years) when the bans were implemented. The effects are statistically significant and increase over time: cessation rates of 10% by 2011, 25% by 2012, and nearly 28% by 2013. Conversely, we find no significant effects of the bans on cessation among individuals who had been smoking for at least four years as of 2009.

Although this finding remains consistent when controlling for age squared (Online Appendix Table B.9), it may suggest an alternative interpretation: younger individuals with lower levels of addiction might frequent public spaces more often, leading to greater exposure to the bans. To explore this further, we consider a flexible specification in Online Appendix Table B.10, where the treatment interacts with indicators for different levels of addiction (1, 2, . . . , 5, and 6 or more years). The results indicate that the effects of smoking bans are significantly stronger among individuals with addiction levels of up to 3 years, compared to those with 4, 5, or 6 or more years, particularly after accounting for age squared. This observation supports our interpretation based on the theory of rational addiction.

In conclusion, when addiction levels are still low, the cost imposed by smoking bans may raise the marginal cost of current consumption beyond the marginal benefit from addiction reinforcement, prompting cessation among young adults. In contrast, for individuals with more established addictions, the additional cost imposed by the bans may not be sufficient to outweigh the reinforcing effects of addiction, particularly for those who are more present-oriented.

5.2. Robustness

In this paper, we focused on the cohort treated in 2009. However, in Online Appendix Table B.4 we show that our results are very similar when also considering two additional capitals treated in 2010 and 2011 under a staggered design, where average effects are obtained according to Sun and Abraham (2021). This result is expected, as the 2009 cohort includes twelve of the fourteen treated capitals and accounts for more than 95% of the weight in every period.

Our estimates are based on unbalanced samples of young adults aged 15 to 29 in each year of analysis. As shown in Online Appendix Table B.12, our results remain robust when using a balanced sample of young adults defined by birth cohort.

We additionally performed sensitivity analyses to address potential confounding factors, such as socioeconomic status, which could influence smoking behavior and survey response rates. In Online Appendix Table B.13, we present the effects of smoking bans by enforcement level, conditional on household income per capita. Furthermore, we split the sample into income groups below and above the minimum wage to examine whether our results are driven by any specific group. Overall, we find that heterogeneity by enforcement levels seems to be more important than heterogeneity by income. If anything, the effects are stronger and more immediate among lower-income individuals (again, only for highly-enforced bans). Social norms are therefore unlikely to be driving results for high-income individuals. We conclude that our estimates can be attributed to the smoking bans and variation in enforcement, rather than sample composition or response rate biases.

In order to verify whether our findings are explained by state- or capital-level bans, we regressed smoking prevalence on the treatment variable interacted with the level of enforcement (low or high) and with the unit of implementation (capital or state). The results can be found in the Online Appendix Table B.14, and are overall robust: in both cases, smoking bans reduced smoking prevalence when strongly enforced. Since state-level smoking bans are likely exogenous to smoking prevalence among young adults within capitals, this finding supports the causal interpretation of our estimates.

Finally, we also estimate the effects of restricting the sample to five treated and five untreated capitals in the Central-Southern region of Brazil, such that they are similar in socio-economic characteristics. The results are presented in Online Appendix Table B.15, and are consistent with our main findings. Although we lose power due to a smaller sample size, the estimates are consistent with smoking bans reducing prevalence mostly by cessation. When restricting to capitals located in the North of Brazil, we find no effects of weakly enforced smoking bans.

6. Conclusion

This paper evaluated the impacts of smoking bans on smoking prevalence, initiation, and cessation among young adults. We explored the regional and timing variations in the implementation of comprehensive smoking bans across Brazilian capitals from 2009 to 2013. For that, we employed a generalized difference-in-differences design to estimate heterogeneous impacts by the time of exposure to the restrictions and their level of enforcement. Our paper is the first to use nationally representative longitudinal data on cigarette consumption in Brazil. To the best of our knowledge, we are among the first studies to provide a broad assessment of the dynamic effects of smoking bans in the context of a developing country, where policy enforcement is especially relevant.

Our estimates suggest that smoking bans reduced smoking prevalence among young adults by 18% after four years of exposure. These effects are mainly driven by strongly enforced bans, and, in general, no significant impacts were found in the cities where the bans were weakly enforced. This effect represents more than 147 thousand young individuals in the cohort treated by local smoking bans in 2009, which can be translated to an avoided cost of USD 86.6 million in 2015 values, considering the direct healthcare costs of USD 588 per smoker in Brazil (Pinto et al., 2019). Furthermore, the impacts were driven by smoking cessation, in particular among young adults who had low levels of addiction when the bans were implemented. For this group, cessation increased by more than 28% after four years of exposure to the bans. Therefore, more than 90,000 young individuals have quit cigarette use due to local smoking bans implemented in Brazil, which resulted in USD 53 million of avoided healthcare costs in 2015 values.

We have also shown the robustness of our results through various sensitivity tests and have ruled out the influence of other tobacco control policies. Therefore, we have provided compelling evidence that public policies like smoking bans can effectively reduce smoking prevalence among young adults, particularly those who have low levels of addiction. However, initiation patterns suggest that their capacity to circumvent legal restrictions may pose a challenge to preventing smoking addiction. Overall, this paper stresses the importance of considering enforcement in policy design, implementation, and empirical evaluation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve language in specific sentences. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Camila Steffens: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Software. **Paula Carvalho Pereda:** Writing – review & editing, Formal analysis, Conceptualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jdeveco.2024.103442.

Data availability

The code and data are available through Zenodo, DOI: 10.5281/zenodo.14310375.

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